



LEARNING OBJECTIVES

After reading this chapter, you should be able to:

1. Identify the components of a pacemaker system.
2. Discuss the terms *triggering*, *inhibition*, *pacing*, *capture*, *electrical capture*, *mechanical capture*, and *sensitivity*.
3. Describe the appearance of a typical pacemaker spike on the electrocardiogram (ECG).
4. Describe the appearance of the waveform on the ECG produced as a result of atrial pacing and ventricular pacing.
5. Explain the differences between single-chamber and dual-chamber pacemakers and between fixed-rate and demand pacemakers.
6. List three types of pacemaker malfunction.
7. Describe how to analyze pacemaker function on the ECG.

KEY TERMS

atrioventricular (AV) interval: In dual-chamber pacing, the length of time between an atrial sensed or atrial paced event and the delivery of a ventricular pacing stimulus; analogous to the PR interval of intrinsic waveforms; also called the *artificial* or *electronic PR interval*

base rate: Rate at which the pacemaker's pulse generator initiates impulses when no intrinsic activity is detected; expressed in pulses per minute (ppm)

capture: The successful conduction of an artificial pacemaker's impulse through the myocardium, resulting in depolarization

demand pacemaker: Pacemaker that discharges only when the patient's heart rate drops below the preset rate for the pacemaker; also known as a *synchronous* or *noncompetitive pacemaker*

dual-chamber pacemaker: Pacemaker that stimulates the atrium and ventricle; dual-chamber pacing is also called *physiologic pacing*

escape interval: Time measured between a sensed cardiac event and the next pacemaker output

failure to capture: A pacemaker malfunction that occurs when the artificial pacemaker stimulus is unable to depolarize the myocardium

failure to pace: A pacemaker malfunction that occurs when the pacemaker fails to deliver an electrical stimulus at its programmed time; also referred to as *failure to fire* or *failure of pulse generation*

fixed-rate pacemaker: Pacemaker that continuously discharges at a preset rate regardless of the patient's intrinsic activity; also known as an *asynchronous pacemaker*

inhibition: Pacemaker response in which the output pulse is suppressed when an intrinsic event is sensed

oversensing: A pacemaker malfunction that results from inappropriate sensing of extraneous electrical signals

paced interval: Period between two consecutive paced events in the same cardiac chamber; also known as the *automatic interval*

pacemaker: A battery-powered device that delivers an electrical current to the heart to stimulate depolarization

sensitivity: The extent to which an artificial pacemaker recognizes intrinsic cardiac electrical activity

threshold: The minimum amount of voltage (i.e., milliamperes) needed to obtain consistent capture

undersensing: A pacemaker malfunction that occurs when the artificial pacemaker fails to recognize spontaneous myocardial depolarization

PACEMAKER SYSTEMS

[Objectives 1, 2]

A cardiac **pacemaker** is a battery-powered device that delivers an electrical current to the heart to stimulate depolarization. A pacemaker system consists of a *pulse generator* (i.e., the power source) and pacing leads. The pulse generator houses a battery and electronic circuitry. The circuitry works like a computer, converting energy from the battery into electrical pulses. Whereas a lithium battery is usually the power source for implanted pacemakers and implantable cardioverter-defibrillators (ICDs), a 9-volt alkaline battery is usually used to power a temporary external pulse generator. A *pacing lead* is an insulated wire that is used to carry an electrical impulse from the pulse generator to the patient's heart. It also carries information about the heart's electrical activity back to the pacemaker. The pacemaker responds to the information received either by sending a pacing impulse to the heart (i.e., *triggering*) or by not sending a pacing impulse to the heart (i.e., **inhibition**).

An artificial pacemaker can be external (a temporary intervention) or implanted. An external pacemaker may be used to control transient disturbances in heart rate or conduction that result from drug toxicity or that occurs during a myocardial infarction (MI) or after cardiac surgery when increased vagal tone is often present. Patients who have chronic dysrhythmias that are unresponsive to medication therapy and that result in decreased cardiac output may require the surgical implantation of a permanent pacemaker or an ICD.

Permanent Pacemakers and Implantable Cardioverter-Defibrillators

A permanent pacemaker is used to treat disorders of the sinoatrial (SA) node (e.g., bradycardias), disorders of the atrioventricular (AV) conduction pathways (e.g., second-degree AV block type II, third-degree AV block), or both, that produce signs and symptoms as a result of inadequate cardiac output (Box 8.1). The pacemaker's pulse generator is usually implanted under local anesthesia into the subcutaneous tissue of the anterior chest just below the right or left clavicle (Fig. 8.1). The patient's handedness, occupation, and hobbies determine whether the pacemaker is implanted on the right or left side.

Box 8.1

Clinical Indications of Decreased Cardiac Output

- Acutely altered mental status
- Dizziness
- Exercise intolerance
- Fainting
- Fatigue
- Heart failure
- Lightheadedness
- Palpitations
- Pulsations in the neck
- Seizures
- Shortness of breath
- Tightness in the chest

An ICD is a programmable device that can deliver a range of therapies (also called *tiered therapy*), including defibrillation, antitachycardia pacing (i.e., *overdrive pacing*), synchronized cardioversion, and bradycardia pacing, depending on the dysrhythmia detected and how the device is programmed (Fig. 8.2). A physician determines the appropriate ICD therapies for each patient.



Did You Know?

During overdrive pacing, the pacemaker is set to pace at a rate faster than the rate of the tachycardia. After a few seconds, the pacemaker is stopped to allow the return of the heart's intrinsic pacemaker.

When ICDs were initially introduced, they were surgically placed subcutaneously in the left upper quadrant of the patient's abdomen. With improvements in technology, ICDs are now small enough to be surgically placed in the pectoral area. ICDs were initially used for patients with one or more sudden cardiac arrests or drug-refractory ventricular fibrillation or sustained ventricular tachycardia. Today, ICDs are also used for patients who have experienced an extensive MI resulting in decreased cardiac output, patients who have a history of MI with recurrent episodes of unexplained syncope, and children with congenital long-QT syndrome, among other conditions.

The circuitry of pacemakers and ICDs is housed in a hermetically sealed case made of titanium that is airtight and impermeable to fluid. Pacemakers and ICDs store information about the activities of the patient's heart, as well as information about the device itself (e.g., number of dysrhythmia episodes; provoking rhythm; dates of pacing, defibrillation, or both). The stored information is periodically retrieved and reviewed by the patient's physician and, if necessary, changes in the device's settings are made. Lithium batteries are almost exclusively used in modern pacemakers and ICDs. Battery life depends on the number of times therapies



Fig. 8.1 Site of implantation of a permanent pacemaker or implantable cardioverter-defibrillator. The pacemaker is usually implanted in the left pectoral region, but it may be placed elsewhere if necessary. (From Forbes C, Jackson W: *Color atlas and text of clinical medicine*, St. Louis, 2003, Mosby.)

ICD: Tiered Therapy

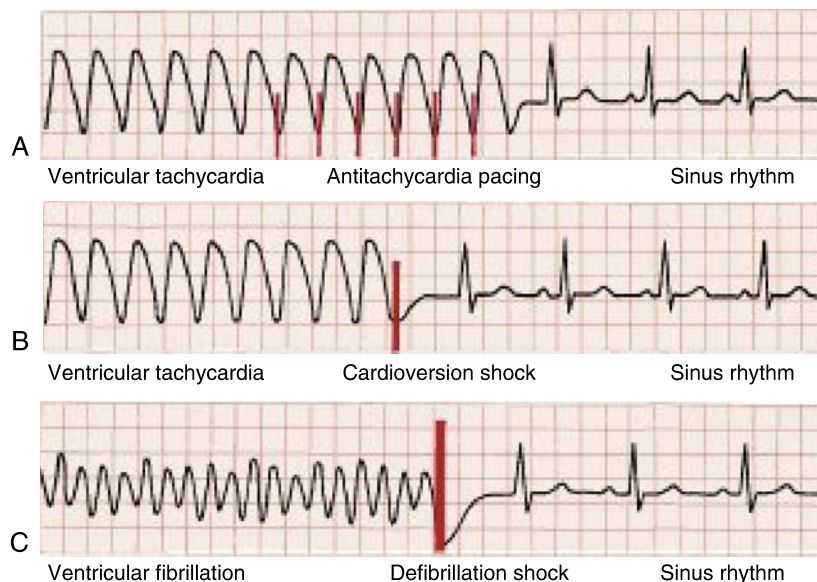


Fig. 8.2 Tiered (staged) dysrhythmia therapy and implantable cardioverter-defibrillators (ICDs). These devices are capable of automatically delivering staged therapy in treating ventricular tachycardia (VT) or ventricular fibrillation (VF), including anti-tachycardia pacing A, or cardioversion shocks B, for VT and defibrillation shocks C, for VF. (From Goldberger AL, Goldberger ZD, Shvilkin A: *Goldberger's clinical electrocardiography: a simplified approach*, ed 8, Philadelphia, 2013, Saunders.)

are delivered, the frequency of pacing, and the number of cardiac chambers paced.

Complications of permanent pacing associated with the implantation procedure include bleeding, local tissue reaction, pneumothorax, cardiac dysrhythmias, air embolism, and thrombosis. Long-term complications of permanent pacing may include infection, electrode displacement, heart failure, fracture of the pacing lead, pacemaker-induced dysrhythmias, externalization of the pacemaker generator, and perforation of the right ventricle with or without pericardial tamponade.

Temporary Pacemakers

The pulse generator of a temporary pacemaker is located externally. Temporary pacing can be accomplished through transvenous, epicardial, or transcutaneous means.

TRANSVENOUS PACING

Transvenous pacemakers stimulate the endocardium of the right atrium or ventricle (or both) by means of an electrode introduced into a central vein, such as the subclavian, femoral, brachial, internal jugular, or external jugular vein. Complications of temporary transvenous pacing include bleeding; infection; pneumothorax; cardiac dysrhythmias; MI; lead displacement; fracture of the pacing lead; hematoma at the insertion site; perforation of the right ventricle with or without pericardial tamponade; and perforation of the inferior vena cava, pulmonary artery, or coronary arteries because of improper placement of the pacing lead.

EPICARDIAL PACING

Epicardial pacing is the placement of pacing leads directly onto or through the epicardium. Epicardial leads may be used when a patient is undergoing cardiac surgery and the outer surface of the heart is easy to reach.

TRANSCUTANEOUS PACING

Transcutaneous pacing (TCP), also called *temporary external pacing* or *noninvasive pacing*, is the use of electrical stimulation through pacing pads positioned on a patient's torso to stimulate contraction of the heart. TCP is indicated for significant bradycardias that are unresponsive to atropine therapy or when atropine is not immediately available or indicated. It may also be used as a bridge until transvenous pacing can be accomplished or the cause of the bradycardia is reversed (e.g., drug overdose, hyperkalemia). *Standby pacing* refers to the application of the pacing pads to the patient's chest in anticipation of possible use, but pacing is not yet needed. For example, standby pacing is often warranted when second-degree AV block type II or third-degree AV block is present in the setting of acute MI.

TCP requires attaching two pacing electrodes to the skin surface of the patient's outer chest wall. The electrical signal exits from the negative terminal on the machine (and subsequently the negative electrode) and passes through the chest wall to the heart. The range of output current of a transcutaneous pacemaker varies depending on the manufacturer (Fig. 8.3).

The primary limitation of TCP is patient discomfort that is proportional to the intensity of skeletal muscle contraction



Fig. 8.3 An example of a monitor/defibrillator that provides cardiac monitoring, defibrillation, cardioversion, and transcutaneous pacing. (Copyright ©2016 Medtronic. All rights reserved. Used with the permission of Medtronic.)

and the direct electrical stimulation of cutaneous nerves. Because TCP is uncomfortable, the administration of sedatives or analgesics is usually necessary in responsive patients. Temporary transvenous pacing is indicated when prolonged TCP is needed.

Possible complications of TCP include the following:

- Coughing
- Skin burns
- Interference with sensing from patient agitation or muscle contractions
- Discomfort as a result of the electrical stimulation of the skin and muscles
- Failure to recognize that the pacemaker is not capturing
- Tissue damage, including third-degree burns, with improper or prolonged TCP
- When pacing is prolonged, pacing **threshold** changes, thereby leading to capture failure

Pacing Lead Systems

[Objectives 2, 3]

Pacemaker lead systems may consist of single, double, or multiple leads. The exposed portion of the pacing lead—that is, the *electrode*—is placed in direct contact with the heart. Pacing, also called *pacemaker firing*, occurs when the pacemaker's pulse generator delivers energy (milliamperes [mA]) through the pacing electrode to the myocardium. Evidence of pacing can be seen as a vertical line or spike on the electrocardiogram (ECG). **Capture** is the successful conduction of an artificial pacemaker's impulse through the myocardium, resulting in depolarization. Capture is obtained after the pacemaker electrode is properly positioned in the heart; with one-to-one capture, each pacing stimulus results in depolarization of the appropriate chamber. On the ECG, evidence of *electrical capture* can be seen as a pacemaker spike followed by an atrial or a ventricular complex, depending on the cardiac chamber that is being paced. *Mechanical capture* is assessed by palpating the patient's pulse or by observing



Fig. 8.4 Example of a temporary unipolar pacing lead. (Copyright ©2016 Medtronic. All rights reserved. Used with the permission of Medtronic.)

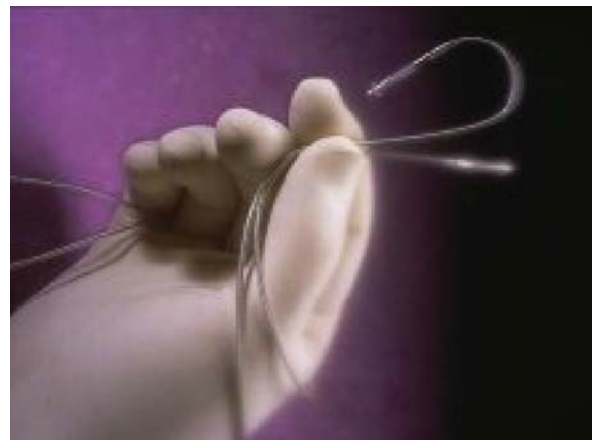


Fig. 8.5 Example of a bipolar pacing lead. (Copyright ©2016 Medtronic. All rights reserved. Used with the permission of Medtronic.)

right atrial pressure, left atrial pressure, or pulmonary artery or arterial pressure waveforms.

A unipolar electrode has one pacing electrode that is located at its distal tip (Fig. 8.4). The negative electrode is in contact with the cardiac tissue, and the pulse generator (located outside the heart) functions as the positive electrode. The pacemaker spike produced by a unipolar lead system is often large because of the distance between the positive and negative electrode. Unipolar leads are less commonly used than bipolar lead systems because of the potential for pacing the chest wall muscles and the susceptibility of the unipolar leads to electromagnetic interference.

A bipolar lead system contains a positive and negative electrode at the distal tip of the pacing lead wire (Fig. 8.5). Most temporary transvenous pacemakers use a bipolar lead system. A permanent pacemaker may have either a bipolar or a unipolar lead system. The pacemaker spike produced by a bipolar lead system is smaller than that of a unipolar system because of the shorter distance between the positive and negative electrodes (Fig. 8.6).

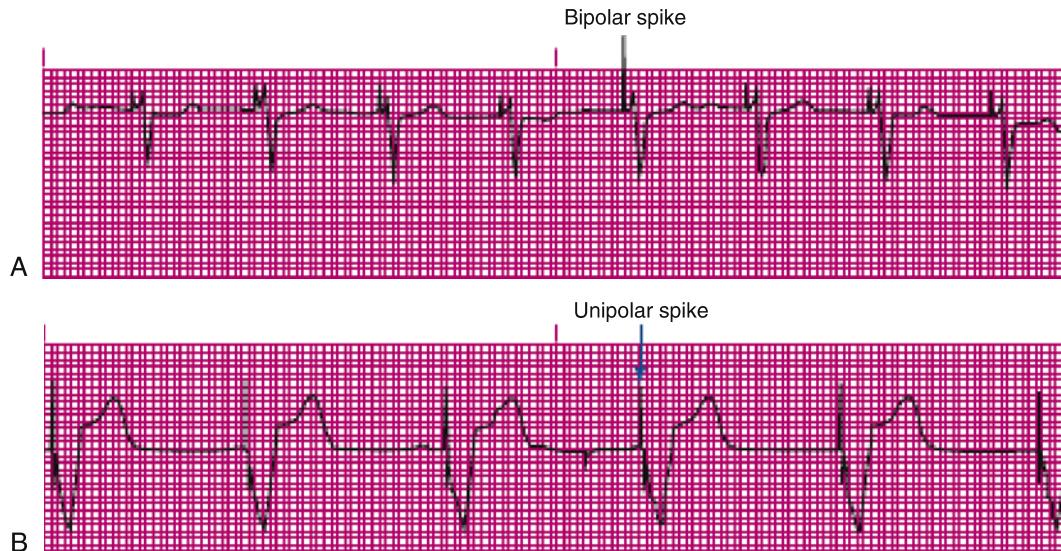


Fig. 8.6 Bipolar and unipolar pacing. A, Pacemaker spike produced by a bipolar lead system. B, Pacemaker spike produced by a unipolar lead system. (From Urden LD, Stacy KM, Lough ME: *Critical care nursing*, ed 8, St. Louis, 2018, Mosby.)

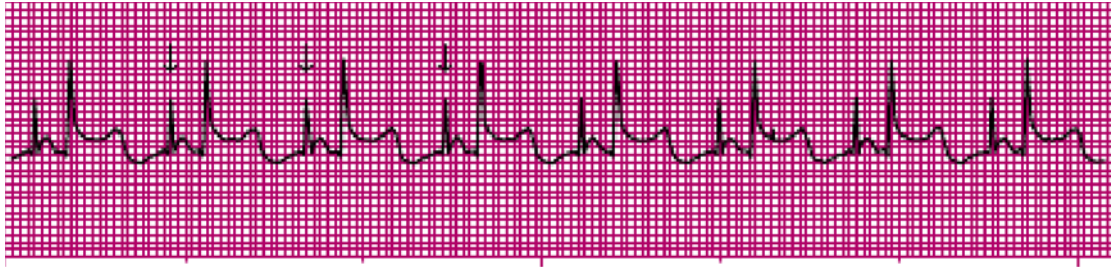


Fig. 8.7 Electrocardiogram of a single-chamber pacemaker with atrial pacing spikes (arrows).

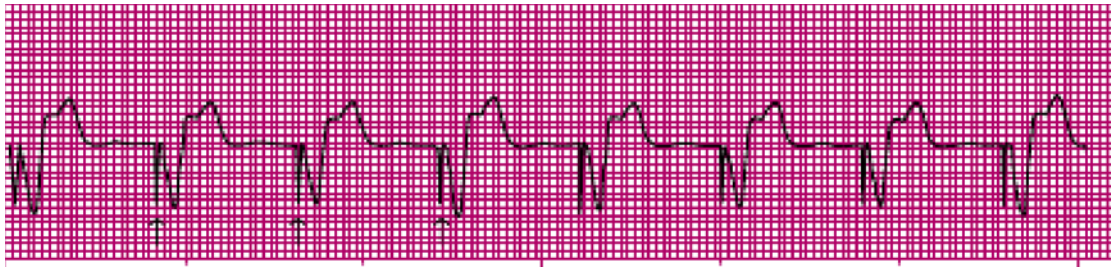


Fig. 8.8 Electrocardiogram of a single-chamber pacemaker with ventricular pacing spikes (arrows).

PACING CHAMBERS AND MODES

A pacemaker system may be single or dual chamber.

Single-Chamber Pacemakers

[Objectives 4, 5]

A pacemaker that paces a single heart chamber, either the atrium or ventricle, has one lead placed in the heart. Atrial pacing is achieved by placing the pacing electrode in the right atrium. Stimulation of the atria produces a pacemaker

spike on the ECG followed by a P wave (Fig. 8.7). Atrial pacing may be used when the SA node is diseased or damaged, but conduction through the AV junction and ventricles is normal. This type of pacemaker is ineffective if an AV block develops because it cannot pace the ventricles.

Ventricular pacing is accomplished by placing the pacing electrode in the right ventricle. Stimulation of the ventricles produces a pacemaker spike on the ECG followed by a wide QRS, resembling a ventricular ectopic beat (Fig. 8.8). The QRS complex is wide because a paced impulse does not follow the normal conduction pathway in the heart. A single-chamber ventricular pacemaker can pace the ventricles but it cannot coordinate pacing with the patient's intrinsic atrial

rate. This results in asynchronous contraction of the atrium and ventricle (i.e., AV asynchrony). Because of this loss of AV synchrony, a ventricular demand pacemaker is rarely used in patients with an intact SA node. Conversely, a ventricular demand pacemaker may be used for patients with chronic atrial fibrillation.

Dual-Chamber Pacemakers

[Objectives 4, 5]

A **dual-chamber pacemaker** uses two leads: One lead is placed in the right atrium and the other in the right ventricle (Fig. 8.9). Dual-chamber pacing is also called *physiologic pacing*. A dual-chamber pacemaker stimulates the right atrium and right ventricle sequentially (stimulating first the atrium and then the ventricle), mimicking normal cardiac physiology and thus preserving the atrial contribution to ventricular filling (i.e., atrial kick).

When spontaneous atrial depolarization does not occur within a preset interval, the atrial pulse generator fires and stimulates atrial depolarization at a preset rate. The pacemaker is programmed to wait—simulating the normal delay in conduction through the AV node (i.e., the PR interval). The artificial or electronic PR interval is referred to as an **atrioventricular (AV) interval** (Fig. 8.10). If spontaneous

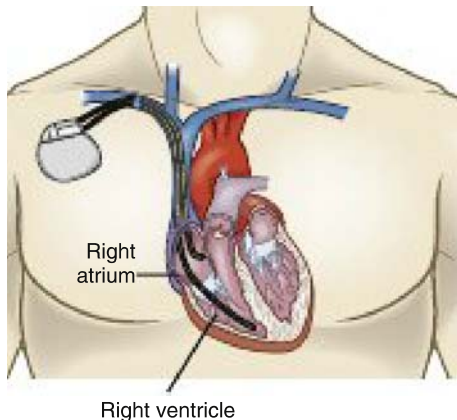


Fig. 8.9 Pacing leads in both the atrium and the ventricle enable a dual-chamber pacemaker to sense and pace in both heart chambers. (From Lewis SL, Bucher L, Heitkemper MM, Harding MM: *Medical-surgical nursing: assessment and management of clinical problems*, ed 10, St. Louis, 2017, Elsevier.)

ventricular depolarization does not occur within a preset interval, the pacemaker fires and stimulates ventricular depolarization at a preset rate.

Biventricular Pacemakers

A *biventricular pacemaker* has three leads—one lead for each ventricle and one lead for the right atrium. This device uses cardiac resynchronization therapy to restore normal simultaneous ventricular contraction for patients with heart failure, thereby improving cardiac output and exercise tolerance.

Fixed-Rate Pacemakers

[Objective 5]

A **fixed-rate pacemaker**, also known as an *asynchronous pacemaker*, continuously discharges at a preset rate (usually 70 to 80 impulses/min) regardless of the patient's heart rate or metabolic demands. An advantage of the fixed-rate pacemaker is its simple circuitry, reducing the risk of pacemaker failure; however, this type of pacemaker does not sense the patient's own cardiac rhythm. This may result in competition between the patient's cardiac rhythm and that of the pacemaker. Ventricular tachycardia or ventricular fibrillation may be induced if the pacemaker were to fire during the T wave (i.e., the vulnerable period) of a preceding patient beat. Fixed-rate pacemakers are not often used today.

Demand Pacemakers

[Objective 5]

A **demand pacemaker**, also known as a *synchronous or non-competitive pacemaker*, discharges only when the patient's heart rate drops below the pacemaker's **base rate**. For example, if the demand pacemaker was preset at a base rate of 70 impulses/min, it would sense the patient's heart rate and allow electrical impulses to flow from the pacemaker through the pacing lead to stimulate the heart only when the rate falls below 70 beats/min. Demand pacemakers can be programmable or nonprogrammable. The voltage level and impulse rate are preset at the time of manufacture in nonprogrammable pacemakers.

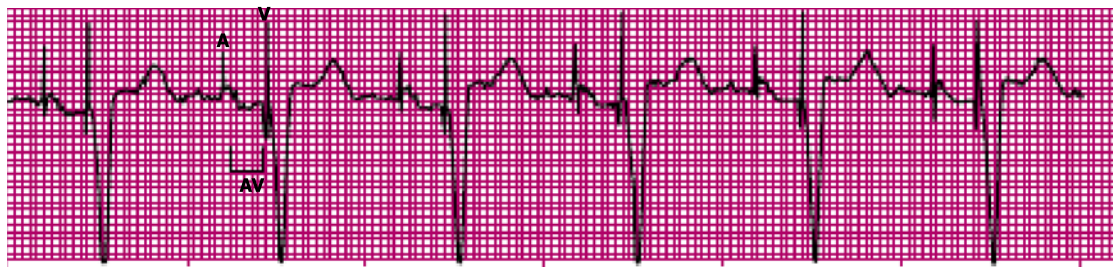


Fig. 8.10 Electrocardiogram of a dual-chamber pacemaker with atrial pacing spikes (A) and ventricular pacing spikes (V). AV, Atrioventricular interval.

TABLE 8.1 Revised NASPE/BPEG Generic Code for Antibradycardia Pacing

Position	I	II	III	IV	V
Category	Chamber(s) Paced	Chamber(s) Sensed	Response to Sensing	Rate Modulation	Multisite Pacing
	O = None A = Atrium V = Ventricle D = Dual (A + V) S = Single (A or V)	O = None A = Atrium V = Ventricle D = Dual (A + V) S = Single (A or V)	O = None T = Triggered I = Inhibited D = Dual (T + I)	O = None R = Rate modulation	O = None A = Atrium V = Ventricle D = Dual (A + V)
Manufacturer's designation only					

NASPE/BPEG, North American Society of Pacing and Electrophysiology/British Pacing and Electrophysiology Group.

From Bernstein A, Daubert J, Fletcher R, Hayes D, et al. The revised NASPE/BPEG generic code for antibradycardia, adaptive-rate, and multisite pacing. North American Society of Pacing and Electrophysiology/British Pacing and Electrophysiology Group, *Pacing Clin Electrophysiol* 2002;25(2):260-264.

Pacemaker Codes

[Objective 5]

Pacemaker codes are used to assist in identifying a pacemaker's preprogrammed pacing, sensing, and response functions (Table 8.1). The *first letter* of the code identifies the heart chamber (or chambers) paced (stimulated). A pacemaker used to pace only a single chamber is represented by either A (atrial) or V (ventricular). A pacemaker capable of pacing in both chambers is represented by D (dual). The *second letter* identifies the chamber of the heart where patient-initiated (i.e., intrinsic) electrical activity is sensed by the pacemaker. The *third letter* indicates how the pacemaker will respond when it senses patient-initiated electrical activity. The *fourth letter* identifies the availability of rate modulation (i.e., the pacemaker's ability to adapt its rate to meet the body's needs caused by increased physical activity and then increase or decrease the pacing rate accordingly). A pacemaker's rate modulation capability may also be referred to as *rate responsiveness* or *rate adaptation*. The *fifth letter* denotes multisite pacing (i.e., biventricular pacing or more than one pacing site in one ventricle). Although the use of all five letters is sometimes needed for completeness, the first three letters are *always* required (Bernstein et al, 2002).

The ventricular demand (VVI) pacemaker is a common type of pacemaker. With this device, the pacemaker electrode is placed in the right ventricle (V); the ventricle is sensed (V) and the pacemaker is inhibited (I) when spontaneous ventricular depolarization occurs within a preset interval. When spontaneous ventricular depolarization does not occur within this preset interval, the pacemaker fires and stimulates ventricular depolarization at a preset rate (Fig. 8.11). P waves can appear anywhere in the cardiac cycle and have no relationship with the QRS complexes because a ventricular demand pacemaker does not sense or pace atrial activity. A disadvantage of ventricular demand pacing is its fixed rate, regardless of the patient's level of physical activity.

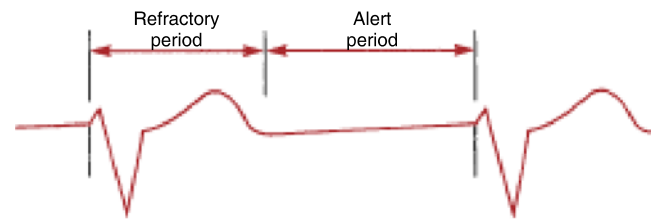


Fig. 8.11 During the refractory period, a ventricular demand pacemaker does not sense any electrical activity. The refractory period is followed by the alert period. If no QRS complex is sensed by the end of the alert period, the pacemaker fires and stimulates ventricular depolarization at a preset rate. (From Goldberger AL: *Clinical electrocardiography: a simplified approach*, ed 7, St. Louis, 2006, Mosby.)

A dual-chamber pacemaker may also be called a *DDD pacemaker*, indicating that both the atrium and ventricle are paced (D), both chambers are sensed (D), and the pacemaker has both a triggered and inhibited mode of response (D) (Fig. 8.12). The presence of a DDD pacemaker does not necessarily mean that the pacemaker is in DDD mode. DDD pacemakers can be programmed to VVI mode, depending on patient need (e.g., the development of chronic atrial fibrillation).

A defibrillator code was developed in 1993, and it is used to describe the capabilities and operation of ICDs (Table 8.2).

PACEMAKER MALFUNCTION

Problems that can occur with pacing include failure to pace, failure to capture, and failure to sense (e.g., undersensing, oversensing).

Failure to Pace

[Objective 6]

Failure to pace, also referred to as *failure to fire* or *failure of pulse generation*, is a pacemaker malfunction that occurs when the pacemaker fails to deliver an electrical stimulus at its programmed time. Failure to pace is recognized on

Dual-Chamber (DDD) Pacemaker Functions



Fig. 8.12 A dual-chamber (DDD) pacemaker senses and paces in both the atria and ventricles. The pacemaker emits a stimulus (spike) whenever an intrinsic P wave or QRS complex is not sensed within some programmed time interval. (From Goldberger AL: *Clinical electrocardiography: a simplified approach*, ed 7, St. Louis, 2006, Mosby.)

TABLE 8.2 NASPE/BPEG Defibrillator Codes

Position I Shock Chamber	Position II Antitachycardia Pacing Chamber	Position III Tachycardia Detection	Position IV Antibradycardia Pacing Chamber
O = None A = Atrium V = Ventricle D = Dual (A + V)	O = None A = Atrium V = Ventricle D = Dual (A + V)	E = Electrogram H = Hemodynamic	O = None A = Atrium V = Ventricle D = Dual (A + V)

NASPE/BPEG, North American Society of Pacing and Electrophysiology/British Pacing and Electrophysiology Group.

From Bernstein AD, Camm AJ, Fisher JD, Fletcher RD, et al. North American Society of Pacing and Electrophysiology policy statement: The NASPE/BPEG defibrillator code, *Pacing Clin Electrophysiol* 1993;16:1776-1780.



Fig. 8.13 Failure to pace. (From Lynn-McHale Wiegand DJ, editor: *AACN procedure manual for critical care*, ed 6, Philadelphia, 2011, Saunders.)

the ECG as an absence of pacemaker spikes, even though the patient's intrinsic rate is less than that of the pacemaker, and a return of the underlying rhythm for which pacing was initiated (Fig. 8.13). Patient signs and symptoms may include bradycardia, chest discomfort, hypotension, and syncope.

Causes of failure to pace are listed in Box 8.2. Treatment may include adjusting the sensitivity setting, replacing the pulse generator battery, replacing the pacing lead, replacing the pulse generator unit, tightening connections between the pacing lead and pulse generator, or removing the source of electromagnetic interference.

Failure to Capture

[Objective 6]

Failure to capture is the inability of the artificial pacemaker stimulus to depolarize the myocardium and is recognized on the ECG by visible pacemaker spikes not followed by P waves (if the electrode is located in the atrium) or QRS

Box 8.2 Causes of Failure to Pace

- Battery failure
- Fracture of the pacing lead wire
- Displacement of the electrode tip
- Pulse generator failure
- Broken or loose connection between the pacing lead and the pulse generator
- Electromagnetic interference
- Sensitivity set too high

complexes (if the electrode is located in the right ventricle) (Fig. 8.14). Patient signs and symptoms may include bradycardia, fatigue, and hypotension.

Causes of failure to capture are shown in Box 8.3. If the problem is a result of low output energy, slowly increasing the output setting (mA) until capture occurs may resolve the problem. With transvenous pacing, repositioning the patient to the left side may promote the contact of a transvenous pacing lead with the endocardium and septum.

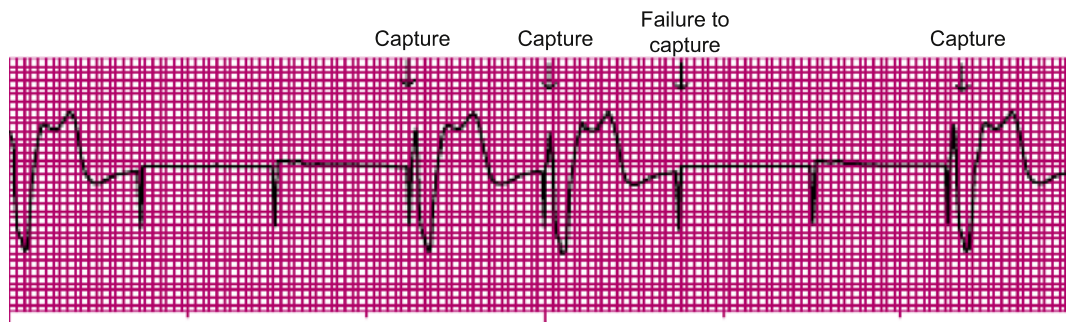


Fig. 8.14 Failure to capture.

Box 8.3 Causes of Failure to Capture

- Displacement of pacing lead wire (common cause)
- Output energy (i.e., mA) set too low (common cause)
- Battery failure
- Fracture of the pacing lead wire
- Edema or scar tissue formation at the electrode tip
- Increased stimulation threshold as a result of medications or electrolyte imbalance

Failure to Sense

[Objective 6]

Sensitivity is the extent to which an artificial pacemaker recognizes intrinsic cardiac electrical activity. **Undersensing** occurs when the artificial pacemaker fails to recognize spontaneous myocardial depolarization and it is recognized on the ECG by pacemaker spikes that occur within P waves, pacemaker spikes that follow too closely behind the patient's QRS complexes, or pacemaker spikes that appear within T waves (Fig. 8.15). Because pacemaker spikes occur when they should not, this type of pacemaker malfunction may result in pacemaker spikes that fall on T waves (i.e., R-on-T phenomenon), competition between the pacemaker and the patient's own cardiac rhythm, or both. The patient may complain of palpitations or skipped beats.

Causes of failure to sense are shown in Box 8.4. Treatment may include increasing the sensitivity setting, replacing the pulse generator battery, or replacing or repositioning the pacing lead.

Oversensing is a pacemaker malfunction that results from inappropriate sensing of extraneous electrical signals. Atrial sensing pacemakers may inappropriately sense ventricular activity; ventricular sensing pacemakers may misidentify a tall, peaked intrinsic T wave as a QRS complex. Oversensing is recognized on the ECG as pacemaker spikes at a rate slower than the pacemaker's preset rate or no paced beats even though the pacemaker's preset rate is greater than the patient's intrinsic rate (Fig. 8.16). Treatment includes adjustment of the pacemaker's sensitivity setting or possible insertion of a bipolar lead if oversensing is caused by unipolar lead dysfunction.

ANALYZING PACEMAKER FUNCTION ON THE ECG

[Objective 7]

To practice analyzing pacemaker function on the ECG, let's look at Fig. 8.17. The first step in our analysis of this rhythm strip should include identification of the patient's underlying rhythm and its rate, if possible. Fig. 8.17 provides a look at the patient's rhythm from two leads, and six QRS complexes are visible in each lead. Each QRS complex is preceded by one upright P wave. Based on this information, we know that the underlying rhythm is sinus in origin. The atrial and ventricular rates are regular at 65 beats/min.

Next, let's look for evidence of paced activity (i.e., atrial pacer spikes, ventricular pacer spikes, or both) and evaluate the paced interval. The **paced interval**, also called the *automatic interval*, is the period between two consecutive paced events in the same cardiac chamber. Measure the distance between two consecutively paced atrial beats using calipers or paper when atrial pacer spikes are present. Because there is no pacemaker spike before any of the P waves in this rhythm strip, there is no evidence of paced atrial activity; however, a pacer spike is clearly visible before each QRS complex.

Because paced ventricular activity is present, we must evaluate the rate and regularity of the ventricular paced interval by measuring the distance between two consecutively paced ventricular beats. The ventricular paced interval is regular at 65 pulses/min. If both atrial and ventricular pacemaker spikes were present, you would know that this patient had a dual-chamber pacemaker. Because only wide-QRS complexes are present and each QRS is preceded by a pacemaker spike, it is reasonable to conclude that this patient has a ventricular pacemaker. Next, compare the escape interval with the paced interval measured earlier. The **escape interval** is the time measured between a sensed cardiac event and the next pacemaker output. The paced interval and escape interval should measure the same. In Fig. 8.17, the paced interval and the escape interval are the same.

Next, analyze the rhythm strip for failure to pace, failure to capture, and failure to sense. Remember that with failure to pace, the ECG will reveal an absence of pacemaker spikes at their programmed time. Because the pacer

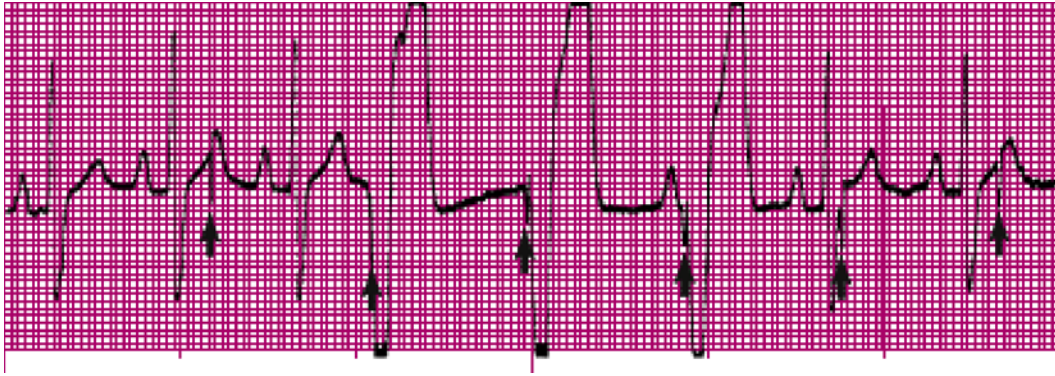


Fig. 8.15 Undersensing.

Box 8.4 Causes of Failure to Sense

- Displacement of the electrode tip (most common cause)
- Battery failure
- Fracture of the pacing lead wire
- Decreased P wave or QRS voltage
- Circuitry dysfunction (e.g., pulse generator unable to process the QRS signal)
- Increased sensing threshold from antiarrhythmic medications
- Severe electrolyte disturbances
- Myocardial perforation

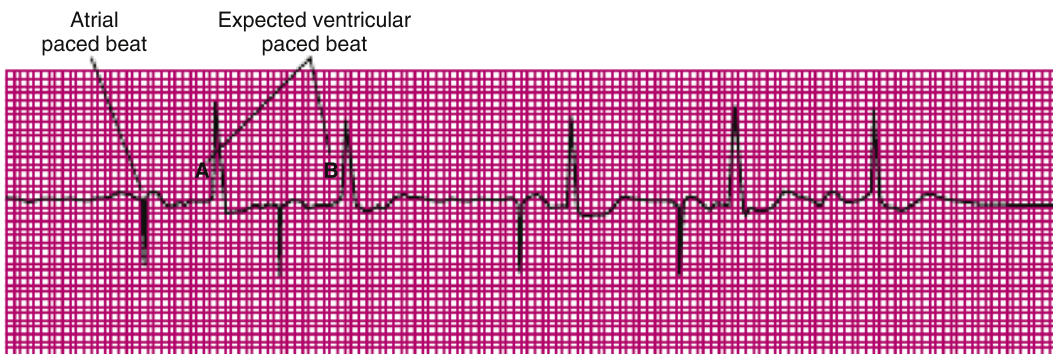


Fig. 8.16 Ventricular oversensing and possibly ventricular pulse generation failure. Ventricular spike expected at 150 msec. Ventricular spike and corresponding ventricular depolarization did not occur at points *A* and *B*. Also, atrial timing reset by oversensed ventricular activity resulted in erratic atrial pacing (suspicious for fracture of ventricular lead). (From Lynn-McHale Wiegand DJ, editor: *AACN procedure manual for critical care*, ed 6, Philadelphia, 2011, Saunders.)

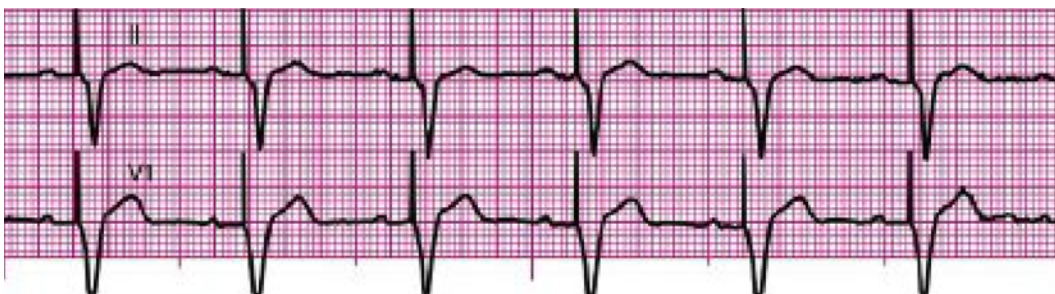


Fig. 8.17 Practice strip, analyzing pacemaker function. (From Aehlert B: *ECG study cards*, St. Louis, 2004, Mosby.)

spikes in Fig. 8.17 occur regularly, we know that failure to pace is not present. When failure to capture occurs, pacer spikes appear regularly but the waveforms after them are periodically absent (i.e., P waves are periodically absent if the pacing electrode is in the atrium, QRS complexes are periodically absent if the pacing electrode is in the ventricle). Because there is a 1:1 relationship between pacemaker spikes and QRS complexes in this rhythm strip, we know that 100% ventricular capture is present. When failure to sense exists, unexpected paced beats or unexpected pacer spikes are present (i.e., undersensing), or prolonged pauses are present (i.e., oversensing). Evaluation of the waveforms and pacer spikes in Fig. 8.17 reveals that there are no unexpected beats, there are no unexpected pacer spikes, and there are no prolonged pauses; thus, failure to sense is not present.

Before we complete our interpretation, we should briefly discuss what looks like ST-segment elevation in this rhythm strip. The QRS complexes are negatively deflected (i.e., a QS configuration), and the ST segments and T waves are in the opposite direction of the last portion of the QRS complex. As you have learned, these findings are common in ventricular rhythms and in left bundle branch block (LBBB). A ventricular paced beat can be thought of as a manmade LBBB. Consider that when LBBB occurs,

the electrical impulse travels down the right bundle branch and depolarizes the right ventricle, and the impulse spreads through the myocardium to depolarize the left ventricle. Pacemakers are most often introduced into the right ventricle. When a pacemaker fires, it sends its impulse into the right ventricle, which depolarizes, and the impulse is spread through the myocardium to depolarize the left ventricle. Therefore, just as with LBBB, ventricular paced rhythms may exhibit ST-segment elevation that is not the result of any infarct-related causes because ventricular depolarization is abnormal.



ECG Pearl

Careful correlation of the patient's ECG, his or her clinical presentation, and the results of other diagnostic studies is essential.

In summary, we will identify the rhythm strip in Fig. 8.17 as a sinus rhythm with a ventricular pacemaker and 100% capture and a paced interval of 65 pulses/min. If the patient's intrinsic rate were different from that of the pacemaker, we would include the paced rate in pulses per minute and the patient's heart rate in beats per minute in our rhythm description.

STOP & REVIEW

Matching

Match the terms below with their descriptions by placing the letter of each correct answer in the space provided.

- | | |
|--|---|
| <ul style="list-style-type: none"> a. AV interval b. Oversensing c. Dual chamber d. Inhibition e. Demand f. Failure to capture g. Base rate h. Pacemaker spike | <ul style="list-style-type: none"> i. Failure to pace j. Rate modulation k. Pulse generator l. Undersensing m. Threshold n. Paced interval o. Fixed rate |
|--|---|

- _____ 1. A vertical line on the ECG that indicates the pacemaker has discharged
- _____ 2. A pacemaker malfunction that occurs when the artificial pacemaker fails to recognize spontaneous myocardial depolarization
- _____ 3. A pacemaker malfunction that occurs when the artificial pacemaker stimulus is unable to depolarize the myocardium
- _____ 4. The period between two consecutive paced events in the same cardiac chamber
- _____ 5. This type of pacemaker stimulates the atrium and ventricle
- _____ 6. In dual-chamber pacing, the length of time between an atrial sensed or atrial paced event and the delivery of a ventricular pacing stimulus; an artificial or electronic PR interval
- _____ 7. This type of pacemaker discharges only when the patient's heart rate drops below the preset rate for the pacemaker
- _____ 8. Ability of a pacemaker to increase the pacing rate in response to physical activity or metabolic demand
- _____ 9. This type of pacemaker continuously discharges at a preset rate regardless of the patient's intrinsic activity
- _____ 10. A pacemaker malfunction that results from inappropriate sensing of extraneous electrical signals
- _____ 11. Power source that houses the battery and circuitry for regulating a pacemaker
- _____ 12. The rate at which the pacemaker's pulse generator initiates impulses when no intrinsic activity is detected; expressed in pulses per minute
- _____ 13. The minimum amount of voltage (i.e., milliamperes) needed to obtain consistent capture
- _____ 14. Pacemaker response in which the output pulse is suppressed when an intrinsic event is sensed
- _____ 15. A pacemaker malfunction that occurs when the pacemaker fails to deliver an electrical stimulus at its programmed time

Pacemaker Rhythms—Practice Rhythm Strips

For each of the following rhythm strips, identify the patient's underlying rhythm (if possible); determine the presence of atrial paced activity, ventricular paced activity, or both; and then analyze the rhythm strip for pacemaker malfunction. All rhythm strips were recorded in lead II unless otherwise noted.

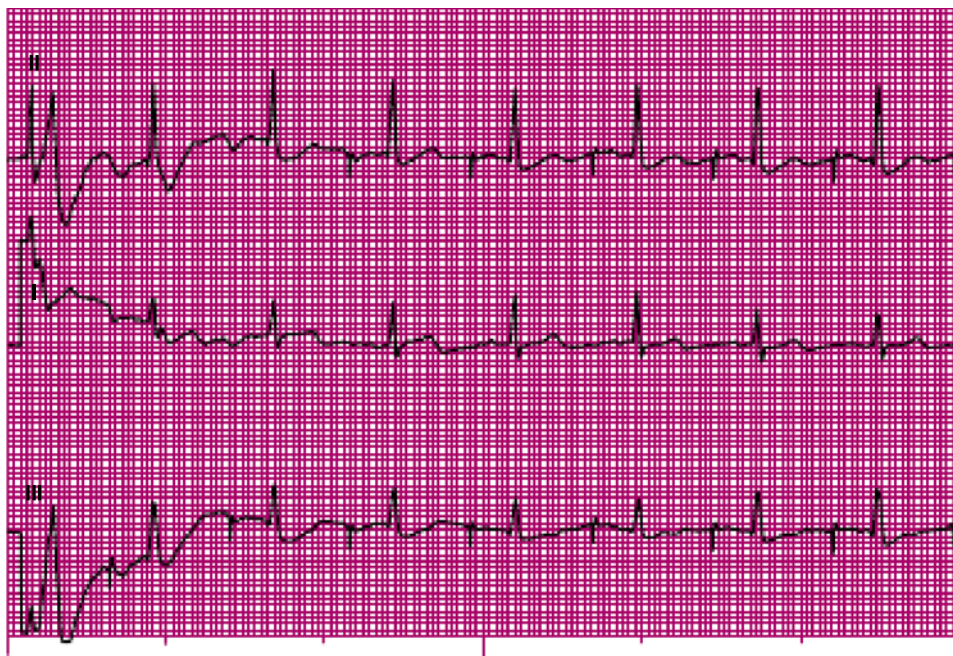


Fig. 8.18 (From Aehlert B: *ECG study cards*, St. Louis, 2004, Mosby.)

16. **Fig. 8.18.** These rhythm strips are from a 52-year-old man with syncope.

Atrial paced activity? _____ Ventricular paced activity? _____

Pacemaker malfunction? _____ Interpretation: _____

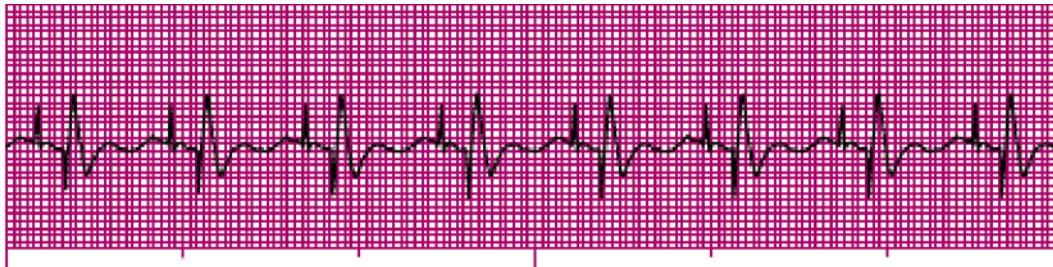


Fig. 8.19 (From Aehlert B: *ECG study cards*, St. Louis, 2004, Mosby.)

17. **Fig. 8.19**

Atrial paced activity? _____ Ventricular paced activity? _____

Pacemaker malfunction? _____ Interpretation: _____

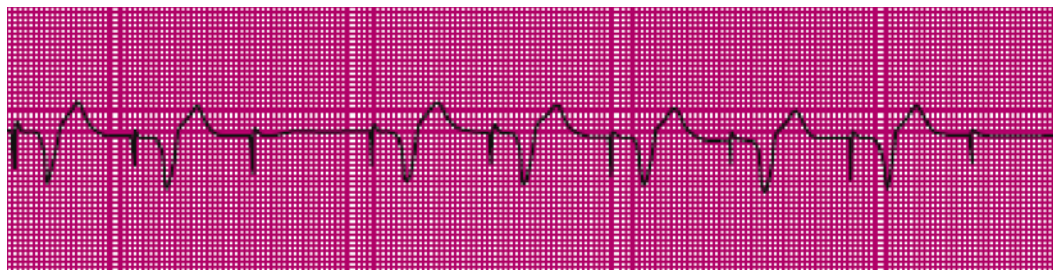


Fig. 8.20 (From Sole ML, Klein DG, Moseley MJ: *Introduction to critical care nursing*, ed 5, Philadelphia, 2008, Saunders.)

18. **Fig. 8.20**

Atrial paced activity? _____ Ventricular paced activity? _____

Pacemaker malfunction? _____ Interpretation: _____

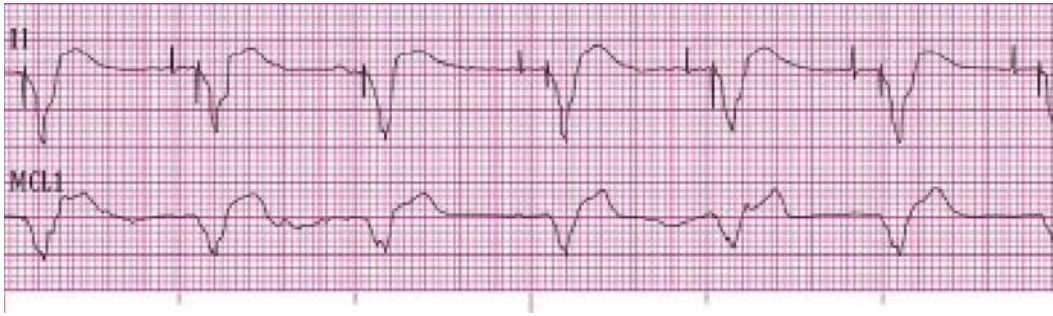


Fig. 8.21 (Modified from Aehlert B: *ECG study cards*, St. Louis, 2004, Mosby.)

19. **Fig. 8.21.** These rhythm strips are from a 63-year-old man complaining of epigastric pain.

Atrial paced activity? _____ Ventricular paced activity? _____

Pacemaker malfunction? _____ Interpretation: _____



Fig. 8.22 (Modified from Aehlert B: *ECG study cards*, St. Louis, 2004, Mosby.)

20. **Fig. 8.22.** This rhythm strip is from a 97-year-old man with chest pain.

Atrial paced activity? _____ Ventricular paced activity? _____

Pacemaker malfunction? _____ Interpretation: _____



Fig. 8.23 (From Aehlert B: *ECG study cards*, St. Louis, 2004, Mosby.)

21. **Fig. 8.23**

Atrial paced activity? _____ Ventricular paced activity? _____

Pacemaker malfunction? _____ Interpretation: _____



Fig. 8.24 (Modified from Aehlert B: *ECG study cards*, St. Louis, 2004, Mosby.)

22. **Fig. 8.24**

Atrial paced activity? _____ Ventricular paced activity? _____

Pacemaker malfunction? _____ Interpretation: _____

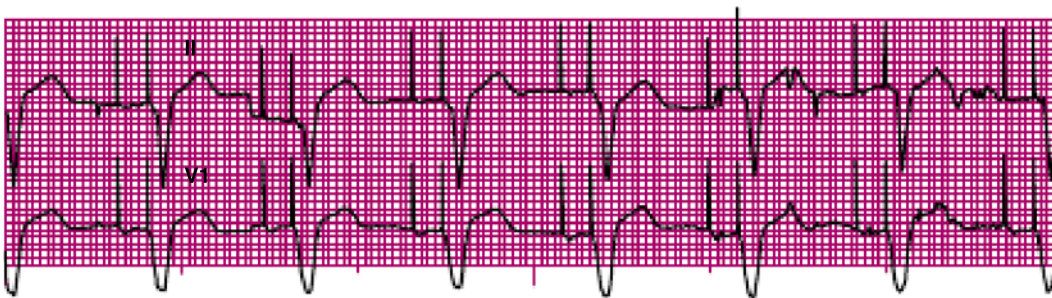


Fig. 8.25 (From Aehlert B: *ECG study cards*, St. Louis, 2004, Mosby.)

23. **Fig. 8.25**

Atrial paced activity? _____ Ventricular paced activity? _____

Pacemaker malfunction? _____ Interpretation: _____

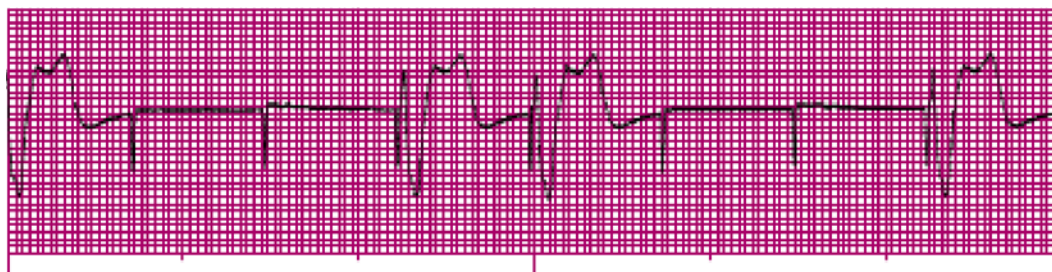


Fig. 8.26

24. **Fig. 8.26**

Atrial paced activity? _____ Ventricular paced activity? _____

Pacemaker malfunction? _____ Interpretation: _____

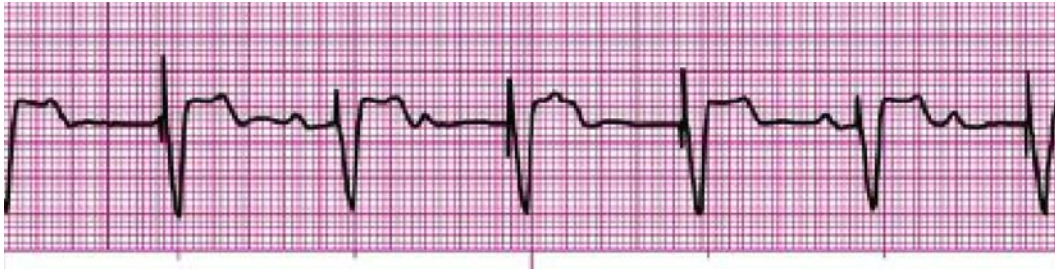


Fig. 8.27 (From Aehlert B: *ECG study cards*, St. Louis, 2004, Mosby.)

25. **Fig. 8.27.** This rhythm strip is from a 90-year-old woman with shortness of breath.

Atrial paced activity? _____ Ventricular paced activity? _____

Pacemaker malfunction? _____ Interpretation: _____

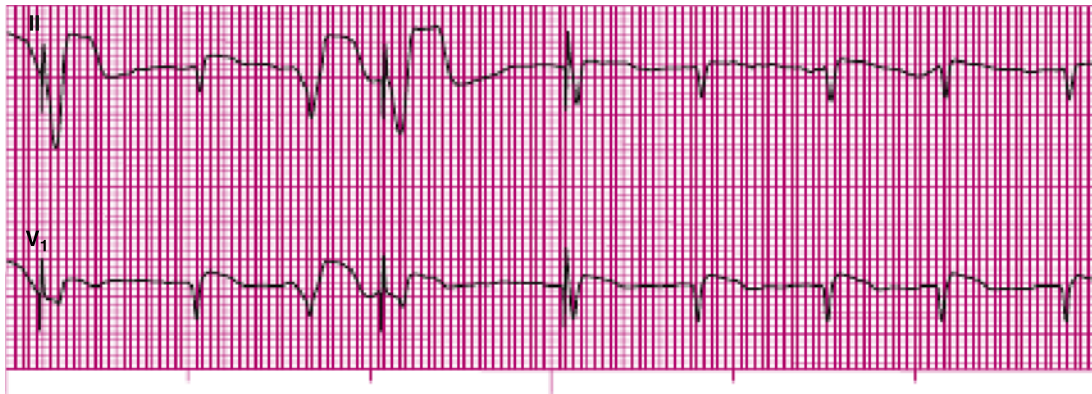


Fig. 8.28

26. **Fig. 8.28**

Atrial paced activity? _____ Ventricular paced activity? _____

Pacemaker malfunction? _____ Interpretation: _____



Fig. 8.29

27. **Fig. 8.29**

Atrial paced activity? _____ Ventricular paced activity? _____

Pacemaker malfunction? _____ Interpretation: _____

**Fig. 8.30**28. **Fig. 8.30**

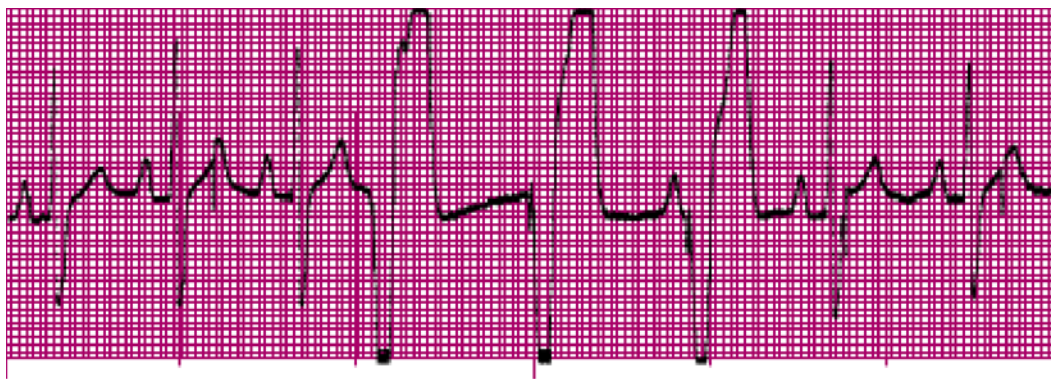
Atrial paced activity? _____ Ventricular paced activity? _____

Pacemaker malfunction? _____ Interpretation: _____

**Fig. 8.31** (Modified from Aehlert B: *ECG study cards*, St. Louis, 2004, Mosby.)29. **Fig. 8.31**. This rhythm strip is from an 80-year-old woman complaining of weakness.

Atrial paced activity? _____ Ventricular paced activity? _____

Pacemaker malfunction? _____ Interpretation: _____

**Fig. 8.32**30. **Fig. 8.32**

Atrial paced activity? _____ Ventricular paced activity? _____

Pacemaker malfunction? _____ Interpretation: _____

STOP & REVIEW / ANSWERS

1. H
2. L
3. F
4. N
5. C
6. A
7. E
8. J
9. O
10. B
11. K
12. G
13. M
14. D
15. I

Practice Rhythm Strip Answers

Note: Because of the distortion of ECGs that can occur during printing, a range of acceptable measurements is provided in the rhythm strip answers throughout this textbook.

16. **Fig. 8.18**
Atrial paced activity? Yes
Ventricular paced activity? No
Pacemaker malfunction? No
Interpretation: Atrial paced rhythm with 100% capture at 79 pulses/min
17. **Fig. 8.19**
Atrial paced activity? Yes
Ventricular paced activity? Yes
Pacemaker malfunction? No
Interpretation: Dual-chamber pacemaker rhythm with 100% capture at 79 pulses/min
18. **Fig. 8.20**
Atrial paced activity? No
Ventricular paced activity? Yes
Pacemaker malfunction? Yes—failure to capture; 7 of 9 paced impulses captured
Interpretation: Ventricular paced rhythm at 65 pulses/min with failure to capture
19. **Fig. 8.21**
Atrial paced activity? Yes
Ventricular paced activity? Yes
Pacemaker malfunction? No
Interpretation: Dual-chamber pacemaker rhythm with 100% capture at 88 pulses/min
20. **Fig. 8.22**
Atrial paced activity? Yes
Ventricular paced activity? Yes
Pacemaker malfunction? Yes—failure to capture; 4 of 6 paced impulses captured
Interpretation: Dual-chamber pacemaker rhythm at 60 pulses/min with failure to capture
21. **Fig. 8.23**
Atrial paced activity? No
Ventricular paced activity? Yes
Pacemaker malfunction? No
Interpretation: Ventricular paced rhythm with 100% capture at 68 pulses/min
22. **Fig. 8.24**
Atrial paced activity? No
Ventricular paced activity? Yes
Pacemaker malfunction? No
Interpretation: Ventricular paced rhythm with 100% capture at 80 pulses/min
23. **Fig. 8.25**
Atrial paced activity? Yes
Ventricular paced activity? Yes
Pacemaker malfunction? No
Interpretation: Dual-chamber pacemaker rhythm with 100% capture at 71 pulses/min
24. **Fig. 8.26**
Atrial paced activity? No
Ventricular paced activity? Yes
Pacemaker malfunction? Yes—failure to capture; 3 of 7 paced impulses captured (the first complex is not counted because the pacer spike before the complex is not visible)
Interpretation: Ventricular paced rhythm at 79 pulses/min with failure to capture
25. **Fig. 8.27**
Atrial paced activity? No
Ventricular paced activity? Yes
Pacemaker malfunction? No
Interpretation: Ventricular-paced rhythm with 100% capture at 60 pulses/min; underlying rhythm appears to be a third-degree AV block
26. **Fig. 8.28**
Atrial paced activity? No
Ventricular paced activity? Yes
Pacemaker malfunction? No
Interpretation: Ventricular demand pacemaker at 60 pulses/min; underlying rhythm appears to be atrial fibrillation
27. **Fig. 8.29**
Atrial paced activity? No
Ventricular paced activity? Yes
Pacemaker malfunction? Yes—failure to capture; 3 of 8 paced impulses captured
Interpretation: Ventricular demand pacemaker rhythm at 72 pulses/min with failure to capture

28. Fig. 8.30

Atrial paced activity? No

Ventricular paced activity? Yes

Pacemaker malfunction? No

Interpretation: Ventricular paced rhythm with 100% capture at 74 pulses/min; underlying rhythm appears to be atrial flutter

29. Fig. 8.31

Atrial paced activity? No

Ventricular paced activity? Yes

Pacemaker malfunction? No

Interpretation: Ventricular paced rhythm with 100% capture at 83 pulses/min

30. Fig. 8.32

Atrial paced activity? No

Ventricular paced activity? Yes

Pacemaker malfunction? Yes—failure to sense (undersensing)

Interpretation: Sinus rhythm at 88 beats/min with a ventricular pacemaker and pacemaker malfunction (undersensing); note the pacer spikes in the T waves of the second and eighth beats from the left

REFERENCES

Bernstein, A. D., Daubert, J. C., Fletcher, R. D., Hayes, D. L., Lüderitz, B., Reynolds, D. W., & Sutton, R. (2002). The revised NASPE/BPEG generic code for antibradycardia, adaptive-rate, and multisite pacing. North American Society of Pacing and Electrophysiology/British Pacing and Electrophysiology Group. *Pacing Clin Electrophysiol*, 25(2), 260–264.